

Лабораторная Работа №4. Первоначальное конфигурирование сети

Администрирование локальных сетей

Боровиков Д.А.

Российский университет дружбы народов им. Патриса Лумумбы, Москва, Россия

- Боровиков Даниил Александрович
- НПИБд-01-22
- Российский университет дружбы народов
- [1132222006@pfur.ru]

Провести подготовительную работу по первоначальной настройке коммутаторов сети.

Размещение коммутаторов и оконечных устройств

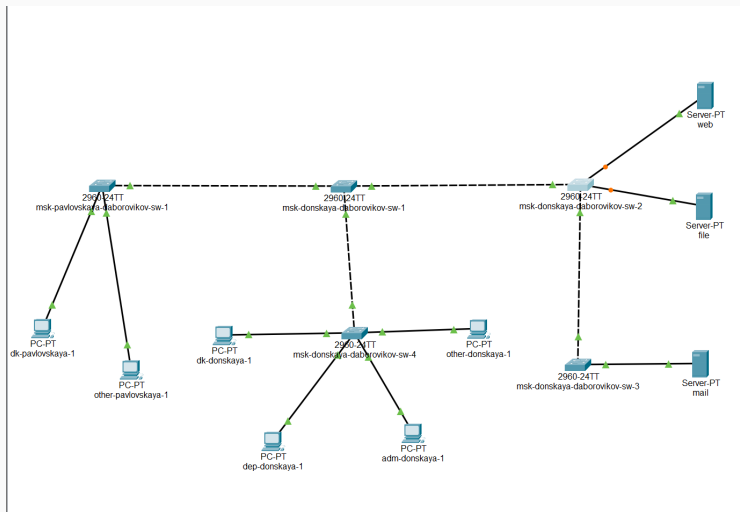
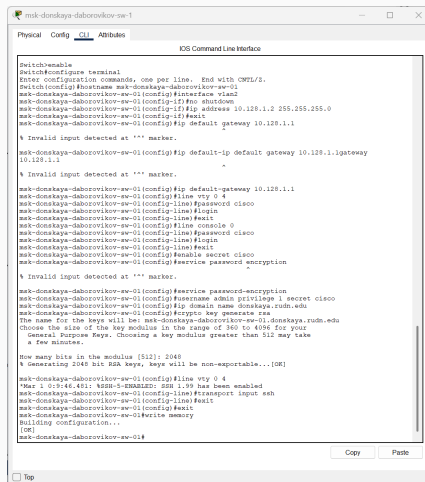


Figure 1: Размещение коммутаторов и оконечных устройств согласно схеме сети L1 4/10

msk-donskaya-daborovikov-sw-1



```
msk-donskaya-daborovikov-sw-1
Physical Config CLI Attributes
IOS Command Line Interface

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CTRL/Z.
Switch(config)#hostname msk-donskaya-daborovikov-sw-01
msk-donskaya-daborovikov-sw-01(config)#interface vlan1
msk-donskaya-daborovikov-sw-01(config-if)#no shutdown
msk-donskaya-daborovikov-sw-01(config-if)#ip address 10.128.1.2 255.255.255.0
msk-donskaya-daborovikov-sw-01(config-if)#exit
msk-donskaya-daborovikov-sw-01(config)#ip default gateway 10.128.1.1

% Invalid input detected at '^' marker.

msk-donskaya-daborovikov-sw-01(config)#ip default-gateway 10.128.1.1
10.128.1.1
% Invalid input detected at '^' marker.

msk-donskaya-daborovikov-sw-01(config)#ip default-gateway 10.128.1.1
msk-donskaya-daborovikov-sw-01(config)#line vty 0 4
msk-donskaya-daborovikov-sw-01(config-line)#password cisco
msk-donskaya-daborovikov-sw-01(config-line)#login
msk-donskaya-daborovikov-sw-01(config-line)#exit
msk-donskaya-daborovikov-sw-01(config)#line console 0
msk-donskaya-daborovikov-sw-01(config-line)#password cisco
msk-donskaya-daborovikov-sw-01(config-line)#login
msk-donskaya-daborovikov-sw-01(config-line)#exit
msk-donskaya-daborovikov-sw-01(config)#enable secret cisco
msk-donskaya-daborovikov-sw-01(config)#service password encryption
% Invalid input detected at '^' marker.

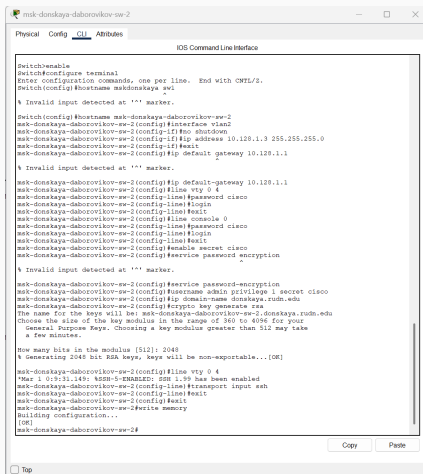
msk-donskaya-daborovikov-sw-01(config)#service password-encryption
msk-donskaya-daborovikov-sw-01(config)#username admin privilege 1 secret cisco
msk-donskaya-daborovikov-sw-01(config)#ip domain name donskeya.rudn.edu
msk-donskaya-daborovikov-sw-01(config)#crypto key generate rsa
The name for the keys will be msk-donskaya-daborovikov-sw-01.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-daborovikov-sw-01(config)#line vty 0 4
*Mar 1 0:39:46.461: %SSH-5-ENABLED: SSH 1.99 has been enabled
msk-donskaya-daborovikov-sw-01(config-line)#transport input ssh
msk-donskaya-daborovikov-sw-01(config-line)#exit
msk-donskaya-daborovikov-sw-01(config)#exit
msk-donskaya-daborovikov-sw-01#write memory
Building configuration...
[OK]
msk-donskaya-daborovikov-sw-01#
```

Figure 2: Первоначальная настройка коммутатора msk-donskaya-daborovikov-sw-1

msk-donskaya-daborovikov-sw-2



```
Switchenable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname mskdonskaya sw1
^
% Invalid input detected at '^' marker.

Switch(config)#hostname msk-donskaya-daborovikov-sw-2
msk-donskaya-daborovikov-sw-2(config)#interface vlan2
msk-donskaya-daborovikov-sw-2(config-if)#no shutdown
msk-donskaya-daborovikov-sw-2(config-if)#ip address 10.128.1.3 255.255.255.0
msk-donskaya-daborovikov-sw-2(config-if)#exit
msk-donskaya-daborovikov-sw-2(config)#ip default gateway 10.128.1.1
^
% Invalid input detected at '^' marker.

msk-donskaya-daborovikov-sw-2(config)#ip default-gateway 10.128.1.1
msk-donskaya-daborovikov-sw-2(config)#line vty 0 4
msk-donskaya-daborovikov-sw-2(config-line)#password cisco
msk-donskaya-daborovikov-sw-2(config-line)#login
msk-donskaya-daborovikov-sw-2(config-line)#exit
msk-donskaya-daborovikov-sw-2(config)#line console 0
msk-donskaya-daborovikov-sw-2(config-line)#password cisco
msk-donskaya-daborovikov-sw-2(config-line)#login
msk-donskaya-daborovikov-sw-2(config-line)#exit
msk-donskaya-daborovikov-sw-2(config)#enable secret cisco
msk-donskaya-daborovikov-sw-2(config)#service password encryption
^
% Invalid input detected at '^' marker.

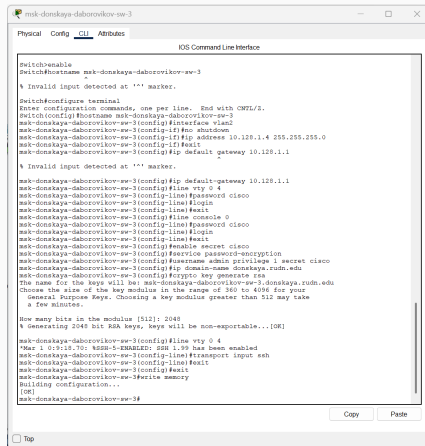
msk-donskaya-daborovikov-sw-2(config)#service password-encryption
msk-donskaya-daborovikov-sw-2(config)#ip domain-name donsokaya.rudn.edu
msk-donskaya-daborovikov-sw-2(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-daborovikov-sw-2.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus (512): 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-daborovikov-sw-2(config)#line vty 0 4
*Mar 1 0:31:14: %SSH-5-ENABLED: SSH 1.99 has been enabled
msk-donskaya-daborovikov-sw-2(config-line)#transport input ssh
msk-donskaya-daborovikov-sw-2(config-line)#exit
msk-donskaya-daborovikov-sw-2(config)#exit
msk-donskaya-daborovikov-sw-2#write memory
Building configuration...
[OK]
msk-donskaya-daborovikov-sw-2#
```

Figure 3: Первоначальная настройка коммутатора msk-donskaya-daborovikov-sw-2

msk-donskaya-daborovikov-sw-3



```
msk-donskaya-daborovikov-sw-3
Physical Config CLI Attributes
IOS Command Line Interface

Switch>enable
Switch#hostname msk-donskaya-daborovikov-sw-3
% Invalid input detected at '^' marker.

Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname msk-donskaya-daborovikov-sw-3
msk-donskaya-daborovikov-sw-3(config)#interface vlan2
msk-donskaya-daborovikov-sw-3(config-if)#no shutdown
msk-donskaya-daborovikov-sw-3(config-if)#ip address 10.128.1.4 255.255.255.0
msk-donskaya-daborovikov-sw-3(config-if)#exit
msk-donskaya-daborovikov-sw-3(config)#ip default-gateway 10.128.1.1
% Invalid input detected at '^' marker.

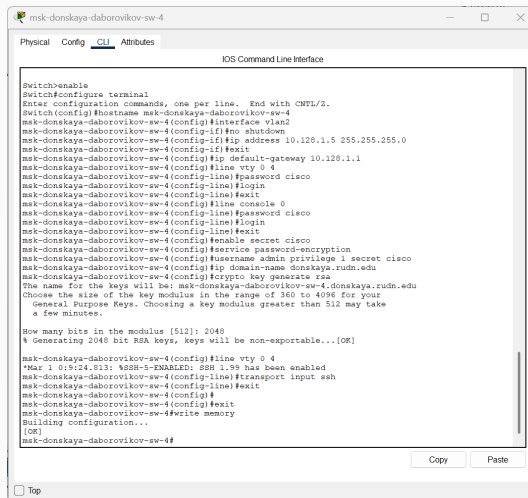
msk-donskaya-daborovikov-sw-3(config)#ip default-gateway 10.128.1.1
msk-donskaya-daborovikov-sw-3(config)#line vty 0 4
msk-donskaya-daborovikov-sw-3(config-line)#password cisco
msk-donskaya-daborovikov-sw-3(config-line)#login
msk-donskaya-daborovikov-sw-3(config-line)#exit
msk-donskaya-daborovikov-sw-3(config)#line console 0
msk-donskaya-daborovikov-sw-3(config-line)#password cisco
msk-donskaya-daborovikov-sw-3(config-line)#login
msk-donskaya-daborovikov-sw-3(config-line)#exit
msk-donskaya-daborovikov-sw-3(config)#enable secret cisco
msk-donskaya-daborovikov-sw-3(config)#service password-encryption
msk-donskaya-daborovikov-sw-3(config)#username admin privilege 1 secret cisco
msk-donskaya-daborovikov-sw-3(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-daborovikov-sw-3(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-daborovikov-sw-3.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-daborovikov-sw-3(config)#line vty 0 4
*Mar 1 0:18:18.70: KERN-5-ENABLED: SSH 1.99 has been enabled
msk-donskaya-daborovikov-sw-3(config-line)#transport input ssh
msk-donskaya-daborovikov-sw-3(config-line)#exit
msk-donskaya-daborovikov-sw-3(config)#exit
msk-donskaya-daborovikov-sw-3#write memory
Building configuration...
[OK]
msk-donskaya-daborovikov-sw-3#
```

Figure 4: Первоначальная настройка коммутатора msk-donskaya-daborovikov-sw-3

msk-donskaya-daborovikov-sw-4



```
msk-donskaya-daborovikov-sw-4
Physical Config CLI Attributes
IOS Command Line Interface

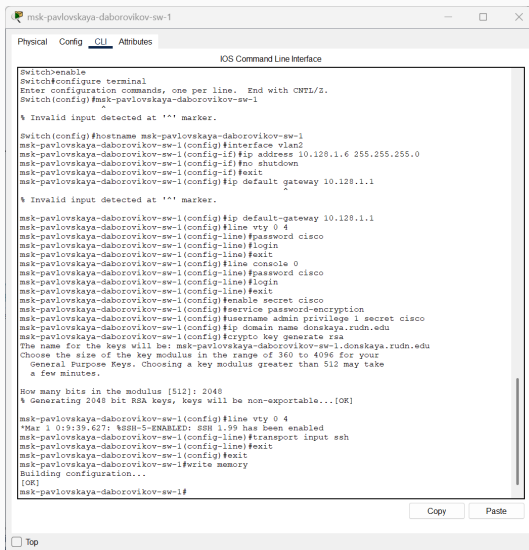
Switch>enable
Switch>configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname msk-donskaya-daborovikov-sw-4
msk-donskaya-daborovikov-sw-4(config)#interface vlan2
msk-donskaya-daborovikov-sw-4(config-if)#no shutdown
msk-donskaya-daborovikov-sw-4(config-if)#ip address 10.128.1.5 255.255.255.0
msk-donskaya-daborovikov-sw-4(config-if)#exit
msk-donskaya-daborovikov-sw-4(config)#ip default-gateway 10.128.1.1
msk-donskaya-daborovikov-sw-4(config)#line vty 0 4
msk-donskaya-daborovikov-sw-4(config-line)#password cisco
msk-donskaya-daborovikov-sw-4(config-line)#login
msk-donskaya-daborovikov-sw-4(config-line)#exit
msk-donskaya-daborovikov-sw-4(config)#line console 0
msk-donskaya-daborovikov-sw-4(config-line)#password cisco
msk-donskaya-daborovikov-sw-4(config-line)#login
msk-donskaya-daborovikov-sw-4(config-line)#exit
msk-donskaya-daborovikov-sw-4(config)#enable secret cisco
msk-donskaya-daborovikov-sw-4(config)#service password-encryption
msk-donskaya-daborovikov-sw-4(config)#username admin privilege 1 secret cisco
msk-donskaya-daborovikov-sw-4(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-daborovikov-sw-4(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-daborovikov-sw-4.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-daborovikov-sw-4(config)#line vty 0 4
*Mar 1 0:9:24.813: %SSH-5-ENABLED: SSH 1.99 has been enabled
msk-donskaya-daborovikov-sw-4(config-line)#transport input ssh
msk-donskaya-daborovikov-sw-4(config-line)#exit
msk-donskaya-daborovikov-sw-4(config)#
msk-donskaya-daborovikov-sw-4(config)#exit
msk-donskaya-daborovikov-sw-4#write memory
Building configuration...
[OK]
msk-donskaya-daborovikov-sw-4#
```

Figure 5: Первоначальная настройка коммутатора msk-donskaya-daborovikov-sw-4

msk-pavlovskaya-daborovikov-sw-1



```
msk-pavlovskaya-daborovikov-sw-1
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#msk-pavlovskaya-daborovikov-sw-1

% Invalid input detected at '^' marker.

Switch(config)#hostname msk-pavlovskaya-daborovikov-sw-1
msk-pavlovskaya-daborovikov-sw-1(config)#interface vlan2
msk-pavlovskaya-daborovikov-sw-1(config-if)#ip address 10.128.1.6 255.255.255.0
msk-pavlovskaya-daborovikov-sw-1(config-if)#no shutdown
msk-pavlovskaya-daborovikov-sw-1(config-if)#exit
msk-pavlovskaya-daborovikov-sw-1(config)#ip default-gateway 10.128.1.1

% Invalid input detected at '^' marker.

msk-pavlovskaya-daborovikov-sw-1(config)#ip default-gateway 10.128.1.1
msk-pavlovskaya-daborovikov-sw-1(config)#line vty 0 4
msk-pavlovskaya-daborovikov-sw-1(config-line)#password cisco
msk-pavlovskaya-daborovikov-sw-1(config-line)#login
msk-pavlovskaya-daborovikov-sw-1(config-line)#exit
msk-pavlovskaya-daborovikov-sw-1(config)#line console 0
msk-pavlovskaya-daborovikov-sw-1(config-line)#password cisco
msk-pavlovskaya-daborovikov-sw-1(config-line)#login
msk-pavlovskaya-daborovikov-sw-1(config-line)#exit
msk-pavlovskaya-daborovikov-sw-1(config)#enable secret cisco
msk-pavlovskaya-daborovikov-sw-1(config)#service password-encryption
msk-pavlovskaya-daborovikov-sw-1(config)#username admin privilege 1 secret cisco
msk-pavlovskaya-daborovikov-sw-1(config)#ip domain-name donskaya.rudn.edu
msk-pavlovskaya-daborovikov-sw-1(config)#crypto key generate rsa
The name for the keys will be: msk-pavlovskaya-daborovikov-sw-1.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

msk-pavlovskaya-daborovikov-sw-1(config)#line vty 0 4
*Mar 1 0:39:27: %SSH-5-ENABLED: SSH 1.99 has been enabled
msk-pavlovskaya-daborovikov-sw-1(config-line)#transport input ssh
msk-pavlovskaya-daborovikov-sw-1(config-line)#exit
msk-pavlovskaya-daborovikov-sw-1(config)#exit
msk-pavlovskaya-daborovikov-sw-1#write memory
Building configuration...
[OK]
msk-pavlovskaya-daborovikov-sw-1#
```

Figure 6: Первоначальная настройка коммутатора msk-pavlovskaya-daborovikov-sw-1

В ходе выполнения лабораторной работы я провел подготовительную работу по первоначальной настройке коммутаторов сети.